

Joshua Fleegler

Berkeley, CA | joshfleegler@gmail.com | 617-999-4092 | [linkedin.com/in/joshua-fleegler](https://www.linkedin.com/in/joshua-fleegler)
josh-fleegler.github.io/Joshua-Fleegler-Engineering-Portfolio/

EDUCATION

UC Berkeley <i>Master's of Engineering, Mechanical Engineering: Product Design concentration</i>	Berkeley, CA May 2027
Tufts University <i>Bachelor's of Science, Mechanical Engineering: Magna Cum Laude, GPA: 3.71</i>	Medford, MA May 2026

EXPERIENCE

Process Engineering Co-op <i>Entegris</i>	January 2025 - June 2025 Bedford MA
- Engineered measurement fixtures using SolidWorks and FDM printing, raising yield of metal filters used in semiconductor manufacturing from 50% to 100% - Collaborated with engineers and operators to iteratively optimize part designs for ergonomics and efficiency, delivering 5–25 iterations per project to balance comfort, function, and manufacturability - Designed a polishing fixture with external manufacturers, cutting one cycle step's time by 90% - Created GD&T drawings for production from external CNC manufacturer and internal reference	
Course Assistant, Mechanics <i>Tufts University</i>	September 2024 - May 2026 Medford, MA
- Ran FEA stress tests for labs, explained difficult concepts to students in office hours - Coordinated with three professors to grade over 100 homework weekly	
Counselor and Ultimate Frisbee Coach <i>URJ Eisner Camp</i>	Summers 2021 - 2024 Great Barrington MA
- Cornerstone Fellowship, one of five counselors selected to attend a national leadership seminar - Created reflective activities for over 600 campers and counselors in coordination with director team	

PROJECTS

Automated Payload Feed System, Space Kinetic (MEng Capstone)	September 2026 - May 2027
- Designing automated reload mechanism for reusable satellite payload deployment system - Conducting engineering and failure mode analysis (FMEA) with CTO and engineering team	
Quadrupedal walking robot	January 2026 - May 2026
- Designed, 3D printed, and assembled 4-legged walking robot in SolidWorks with 2 degrees of freedom per leg - Programmed raspberry pi in python to coordinate 8 motors, motor controller, and IMU	

RESEARCH

Lead Engineering Designer, FIRST STEP (FIRearm Secure sTorage ExPeriment)	September 2025 - Present
- Using SolidWorks and FDM 3D printing to design and produce multiple inert full-size firearm models for gun violence prevention study - Collaborating with academic, clinical, and end-user groups to achieve realistic weight and feel	

SKILLS

CAD and Analysis: SolidWorks, GD&T, FEA
Mechanical Design: Multi-part assemblies, fixture design, design iteration & optimization
Manufacturing: DFM, FDM 3D printing, laser cutting, machine shop tools
Software: Python, MATLAB, SAP, PDM

ACTIVITIES AND LEADERSHIP

Club Ultimate Frisbee	2016 - Present
<i>Needham Public Schools, Tufts (leadership committee), UC Berkeley</i>	
FIRST Robotics: FIRST Tech Challenge and First Lego League <i>Project Manager and Hardware Lead</i>	2014 - 2022
Co-founded and led teams of 20+ as project manager and hardware lead	